METRIC
MIL-PRF-49318A(CR)
19 MAY 1997
SUPERSEDING
MIL-T-49318(CR)
21 November 1991

PERFORMANCE SPECIFICATION
TEST SET, ELECTRONIC SYSTEMS, TS-4348/UV

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification establishes performance, test, manufacture and acceptance requirements for the Test Set, Electronic Systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation. (see 6.2)

DRAWINGS

USA CECOM

A3139775	- Test Set Assy, Electronic Systems TS-4348/UV
A3139812	- Case

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.(see 6.2)

AMERICAN NATIONAL STANDARDS INSTITUTE(ANSI)

ANSI C 64.4-1991	- Methods of Measurement of Radio - Noise Emissions From Low - Voltage Electronic Equipment.

(Application for copies should be addressed to The American National Standards Institute, Inc., 1430 Broadway, N.Y., N.Y. 10018).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

3.2 Materials. The contractor shall select materials that are capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Test Set, Electronic Systems, TS-4348/UV, herein referred to as the "Assessor", consists of the test set, carrying case, shipping/storage case, goggle adapter, driver's viewer adapter, weapon sight adapter, and ancillary equipment.

3.3.1 Purpose. The Assessor is a self-contained test device designed to provide a Go/No-Go check of the night vision systems listed below.

AN/PVS-5 (A, B, C), Night Vision Goggle

AN/PVS-7 (A, B), Night Vision Goggle

AN/AVS-6 (V) 1 & 2, Aviator's Night Vision Imaging System (ANVIS)

AN/PVS-4, Individual Served Weapon Sight

AN/TVS-5, Crew Served Weapon Sight

AN/VVS-2, (V) 1A, 2A, & 3 Viewer, Driver's Night Vision

3.3.2 Design. The Assessor shall meet the dimensions and dimensional tolerances of A3139812 and as specified herein to ensure proper physical interface of equipment built IAW this specification and existing hardware. Inconsistencies between this specification and the drawing shall be decided in favor of this specification.

3.3.3 Weight. The Test Set shall not exceed 400 grams (max) (without batteries or adapters).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.4 Major components. All major components shall conform to the requirements set forth in this performance specification and shall be electronically and mechanically interchangeable (see 3.5.15).

3.5 Performance characteristics.

3.5.1 Light source. A light source equal or similar to a Light Emitting Diode (LED) with a source wavelength of nominally 800nm, ± 25nm shall be provided.

3.5.2 Resolution. The resolution target shall be capable of measuring tube resolution at both high and low light conditions. The resolution target may employ the 1951 US Air Force Resolving Power Test Target (with black bars on a clear background) but shall not use elements 1 through 4 of Group 1. Only one target of this description is required in the test set. Contrast of the target used shall be greater than 99% for both low and high light levels as tested per 4.5.2.

3.5.3 Output collimator.

3.5.3.1 Output collimator diopter. The output Collimator shall present both high and low light resolution patterns at infinity, +0.0/-0.02 diopters when viewed through the test set.

3.5.3.2 Output collimator exit pupil. The output Collimator exit pupil shall be greater than 11mm diameter and less than or equal to 22mm diameter.

3.5.4 Adapters. There shall be no light leakage into the Night Vision System or the Assessor when each of the three adapters are attached to the Assessor. The Driver's Viewer adapter shall present the resolution target image focused at 50 ft ± 10 ft.

3.5.5 Power supply. The power supply shall be a 9.0 V battery. Use only a commercial dry cell battery which is manually replaceable without the use of special tools.

3.5.6 Low battery indicator. Each Assessor shall incorporate a red LED which shall illuminate when battery voltage drops to a level to indicate to the user that the light output (either the high or low light level) is out of the specification range.

3.5.7 Controls and switches. All controls and switches shall be clearly marked per matrix of 3.5.11 to indicate the function of each position. All controls shall be large enough to facilitate operation by the user while wearing arctic mittens. The operation of the Assessor shall not degrade or interfere with the systems under test or with other military equipment. All switches shall be recessed so that they do not protrude past the outer casing of the Assessor body.

3.5.8 Power ON indicator LED. The Assessor shall incorporate a green LED which shall illuminate when the Assessor's selector switch is in either the II or III position. It will be OFF when the switch is in the center position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.5.9 Optical instrument case. The optical instrument case shall accommodate the Assessor in its carrying bag, adapters for all the applicable systems, it is to test, lens tissue paper, a battery and an operator's manual.

3.5.10 Input current. Each Assessor shall require less than 5mA when operated in any of the required test modes and shall not be damaged when subjected to polarity reversal due to inadvertent reversal of the battery. Component operation shall be normal when proper polarity is restored.

3.5.11 Test modes. A total of 4 modes shall be provided by the Assessor according to the following matrix:

Switch Position (Switch II/III)	Switch Position (Switch HIGH/LOW)	
	HIGH	LOW
II (Gen II)	PVS-4, 5, 7 TVS-5, VVS-2 High Light Test	PVS-5, 7 Low Light Test
III (GEN III)	ANVIS High Light Test	ANVIS Low Light Test

3.5.12 Light levels. All light levels shall remain within the specific range throughout the required operational temperature range of +52°C to -32°C. A total of four light levels shall be provided by the Assessor according to the following matrix:

Switch Position (Switch II/III)	Switch Position (Switch HIGH/LOW)	
	On axis output radiance (w/ ($sr*cm^2$))	
	HIGH ± 50 percent	LOW ± 20 percent
II (GEN II) III (GEN III)	$C*8.3*10^{-9}$ $C*3.9*10^{-9}$	$3.41*10^{-11}$ $1.52*10^{-11}$

Where $C = (22mm/ (NVDA\ exit\ pupil\ diameter))^2$

3.5.13 Field of view (FOV). The Assessor shall have a lighted apparent FOV of 30 degrees minimum and shall not be vignetted as observed by the following systems: AN/AVS-6(V)1, (V)2, AN/PVS-4, AN/TVS-5, AN/VVS-2(V) 1A, 2A and 3. The contractor shall accommodate the following fields of view:

40 degree minimum FOV of all required goggles - AN/AVS-6(V)1, (V)2.
15 degree minimum FOV - AN/PVS-4.
9 degree minimum FOV - AN/TVS-5.
45 degree width x 38 degree height FOV - AN/VVS-2 (V) 1A, 2A and 3.

3.5.14 Reliability. The Assessor shall have a minimum Mean Time Between Failure (MTBF) of 10,000 hours. Use of MIL-HDBK-781 is recommended for reliability determination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.5.15 Interchangeability. All units, adapters and replaceable parts shall be physically and functionally interchangeable without modification.

3.6. Environmental conditions.

3.6.1 High temperature. There shall be no degradation of performance or damage when operated at a temperature of 52°C or stored at a temperature of 71°C.

3.6.2 Low temperature. There shall be no degradation of performance or damage when operated at an temperature of -32°C or stored at a temperature of -62°C.

3.6.3 Altitude. There shall be no degradation of performance or damage when operated at an altitude of 15,000 feet or stored at an altitude of 40,000 feet.

3.6.4 Temperature shock. There shall be no degradation of performance when exposed to temperature shock profiles from 71°C to -62°C (non-operating).

3.6.5 Shock. Each Assessor shall withstand shock of 40g's at 11 milliseconds without damage or degradation of performance.

3.6.6 Vibration. There shall be no degradation of performance or damage when exposed to simple harmonic motion having an amplitude of ±0.015 inches (0.030 inch total excursion) with the frequency being varied between 5 and 55 hertz (Hz).

3.6.7 Humidity. The Assessor shall be fully operational with no damage after 10 days exposure up to 65°C and 94, ± 4 percent relative humidity.

3.6.8 Immersion. There shall be no evidence of damage or degradation of performance after 30 minutes exposure at a depth of 3 feet of fresh water.

3.6.9 Fungus. There shall be no evidence of fungus growth nor damage when exposed to fungus growth environments.

3.6.10 Salt fog. There shall be no evidence of damage to any exposed surface or part after exposure to a salt atmosphere.

3.6.11 Sand and dust. There shall be no degradation of performance or damage after exposure to fine sand and dust particles.

3.6.12 Electromagnetic interference (EMI). The Assessor's operational performance shall not be adversely affected during or after exposure to electromagnetic interference at the frequencies between 14.0 KHz and 40.0 GHz for RE02 and RE03 as verified by use of Appendix A. Refer also to ANSI C 64.4 for additional data on EMI measurement methods.

3.7 Treatment and painting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.7.1 Materials and treatments. Materials used in the construction of the Assessor shall be of corrosion-resistant types or shall be processed to resist corrosion by finishing, plating, or similar processes.

3.7.2 Finishes. The finishes used on the Assessor shall possess a non-reflective quality. All external and internal surfaces, except light transmitting elements, that are part of the test light path shall be finished to achieve the lowest reflectance index.

3.8 Nameplates. The outer case shall have a nameplate. The information on the nameplate shall include:

TS-4348/UV
TEST SET, ELEK SYS
MFG IDENT XXXXX
CONT. NO. XXXXXXX
NSN XXXX-XX-XXX-XXXX

3.9 Marking. The inner case shall be marked as follows:

TEST SET
ELECTRONIC SYSTEMS
TS-4348/UV

3.10 Workmanship. All Assessors shall be free from cracks, burrs, chips, dirt, grease or any embedded foreign material. Joints and seams shall be a tight fit and electrical wiring shall be secure with unbroken insulation. All moving parts and adjustments shall be examined to ensure that they move freely throughout their entire range without sticking or binding.

3.11 Environmental stress screening (ESS). Each Assessor shall be exposed to environmental stress screening as established by the contractor and approved by the Government.

3.12 Technical interpretations. The following technical interpretations are, when referenced in Section 3, 4 or 5, mandatory for this specification.

3.12.1 Damage. Damage is defined as an electrical failure or malfunction including arcing, corona, intermittent operation, flickering, brittleness, cracking, corrosion or failure of any finish; condensation, fogging or moisture on any internal surface; dents or breakage of any hardware, connection or component; or any change in alignment.

3.12.2 Room temperature. Room temperature is defined as +23°C ± 10°C.

3.12.3 Infant mortality. Those failures occurring due to poor workmanship or weak parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.12.4 Environmental stress screening (ESS). An ESS failure is any shortcoming, imperfection or operational non-conformance, including a one-time non-repeatable anomaly, either sudden or gradual in nature, which causes the unit under test to deviate from specified limits. ESS failures, include poor workmanship (i.e. solder balls or cracks), defective manufacturing processes (i.e. wave soldering devices which cannot handle the heat) and defective components. Non-ESS failures are those failures directly attributed to improper installation in the test facility, failures of test monitoring equipment, failures occurring during repair, and failures attributed to test equipment generating overstress conditions.

3.12.5 Degradation of performance. A significant change in measurable characteristic (s) which results in the assembly no longer meeting specified requirements or indicates that there is an inherent defect in the operating characteristics of the unit.

3.12.6 Qualification. Unless otherwise specified by the contractor or purchase order, Assessors furnished under this specification shall be products approved as qualified products (see 3.12.7)

3.12.7 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the Project Manager, Night Vision and Electro-Optics or its designated representative. Should the supplier make substantial changes in the process, personnel, location or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified in the contract or purchase order, a number of Assessors shall be provided for first article inspection. This inspection shall include the inspection of 4.2.1 and the tests of 4.2.2.

4.2.1 Inspection. Each Assessor and adapters shall be inspected in accordance with Table I, both in process and when assembled, as required. Presence of one or more defects shall be cause for rejection of the Assessor and /or adapters and be may cause for failure of the first article.

TABLE I. Visual inspection.

Inspection	Requirement paragraph	Inspection Paragraph
Treatment & Painting	3.7	4.7
Nameplate	3.8	4.7
Marking	3.9	4.7
Inspection	Requirement paragraph	Criteria
Workmanship	3.10	4.7
Construction	3.3.2	4.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

Environmental Stress Screening (ESS)	3.11	4.7
Parts and Materials	3.2	4.7

4.2.2 Tests. Assessors with adapters that have passed the visual inspections of 4.2.1 shall be subjected to the inspections in Table II. Twenty (20) units shall be tested per Subgroup 1. At least four (4) sample units passing Subgroup 1 shall be submitted to the interchangeability tests of Subgroup 2 and then one sample of each unit passing interchangeability shall be submitted to the remaining tests of Subgroup 2. Six (6) of the units passing Subgroup 1 shall be subjected to the Humidity tests of Subgroup 3 and ten (10) of the units passing Subgroup 1 shall be subjected to the Reliability tests of Subgroup 3. Unless specified, tests may be performed in any order. Failure of any test or visual inspection shall constitute first article failure.

TABLE II. First article tests.

Inspection	Requirement paragraph	Inspection paragraph	Subgroup 1	2	3
Weight	3.3.3	Visual	X		
Output collimator diopter	3.5.3.1	4.5.3	X		
Output collimator exit pupil	3.5.3.2	4.5.3	X		
Low battery indicator	3.5.6	4.5.6	X		
Power on indicator	3.5.8	4.5.8	X		
Input current	3.5.10	4.5.10	X		
Test modes	3.5.11	4.5.11	X		
Light levels	3.5.12	4.5.1	X		
FOV	3.5.13	4.5.13	X		
Reliability	3.5.14	4.5.14			X
Interchangeability	3.5.15	4.5.15		X	
High temperature	3.6.1	4.6.1	X		
Low temperature	3.6.2	4.6.2	X		
Altitude	3.6.3	4.6.3	X		
Shock	3.6.5	4.6.5	X		
Vibration	3.6.6	4.6.6	X		
Humidity	3.6.7	4.6.7			X
Immersion	3.6.8	4.6.8	X		
Fungus	3.6.9	4.6.9		X	
Salt fog	3.6.10	4.6.10		X	
Sand and dust	3.6.11	4.6.11		X	
EMI	3.6.12	4.8		X	

4.2.3 Disposition of first article samples. Disposition of first article samples shall be as specified in the contract or purchase order (see 6.2.g).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.3 Conformance inspection.

4.3.1 Inspection. Each Assessor and associated adapters shall be inspected to the requirements of Table I in process and when assembled as applicable. Presence of one or more defects shall be cause for rejection of that component or attachment.

4.3.2 Tests.

4.3.2.1 Group A inspection. Each Assessor and adapter shall be tested in accordance with Table III. Each Assessor shall have been subjected to and passed Environmental Stress Screening (ESS) as specified in 3.11. The tests listed Table III may be performed in any order. Failure of an Assessor or adapter to pass any tests shall be cause for rejection of that Assessor or adapter. Units which pass the Group A test shall be formed into lots by the contractor.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Resolution target	3.5.2	4.5.2
Light levels	3.5.1	4.5.1

4.3.2.2 Group B inspection. Assessors and adapters which have successfully passed Group A tests shall be sampled and subjected to the tests prescribed in Table IV. Sampling shall be in accordance with the contract or purchase order (see 6.2.f).

TABLE IV. Group B Inspection.

Inspection	Requirement paragraph	Test paragraph
Low battery Indicator	3.5.6	4.5.6
Output collimator	3.5.3	4.5.3
Weight	3.3.3	Visual

4.3.2.3 Group C inspection. Group C inspection shall be conducted on lots that have successfully passed Group A tests. Sampling shall be in accordance with the contract or purchase order (see 6.2.f).

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Inspection paragraph	Subgroup 1	2
High Temperature	3.6.1	4.6.1	X	
Low Temperature	3.6.2	4.6.2	X	
Altitude	3.6.3	4.6.3	X	
Temperature shock	3.6.4	4.6.4	X	
Shock	3.6.5	4.6.5	X	
Vibration	3.6.6	4.6.6	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

Humidity	3.6.7	4.6.7		X
Immersion	3.6.8	4.6.8	x	

4.3.2.3.1 Group C failure. Action required upon occurrence of a Group C failure shall be as specified in the contract or purchase order (see 6.2.g).

4.3.2.3.2 Disposition of Group C samples. Units completing Group C inspection shall be refurbished, as necessary, subjected to the tests of Table II and shipped as part of the contract.

4.3.2.4 Group D inspection. Group D inspection shall be conducted on units selected from lots which have successfully passed Group B tests.

TABLE VI. Group D inspection.

Inspection	Requirement paragraph	Test method
Reliability	3.5.14	4.5.14

4.4 Reliability test. Reliability testing shall be conducted on Assessor units selected from units which have passed the tests in 4.3.2.1. Tests shall be conducted in accordance with Table VI.

4.4.1 Sampling for reliability testing. The reliability acceptance test shall be performed on Assessor units from each production lot submitted. The sample size shall be two units per lot for a total of ten samples.

4.4.2 Reliability failures. Actions required relative to reliability failures shall be as specified in the contract or purchase order (see 6.2.g).

4.4.3 Reinspection of reliability sample units. Unless otherwise specified, sample units which have been subjected to and passed reliability tests may be accepted on contract or purchase order provided all damage is repaired and the units are re-subjected to and pass Group A inspection.

4.5 Methods of inspection.

4.5.1 Light levels. Each required light level of the Assessor shall be measured by mounting a calibrated photo diode detector assembly in the Goggle adapter and measuring the current through the detector with a digital ammeter. The measured current will be compared to a calibration matrix.

4.5.2 Resolution. The test set shall be mounted in a holding fixture with a high intensity light source positioned at the Assessor’s test port.

4.5.3 Output collimation. The Assessor shall be positioned in a holding fixture with a high intensity light source illuminating the Assessor’s test port. A diopter scope shall be used to measure the exact diopter focus. The reading shall be between 0.0 and -0.02 diopters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.5.4 Adapters. Attach each of three adapters to the Assessor and check for light leakage in accordance with the requirement of 3.5.4.

4.5.5 Power supply. Examine the battery for conformance with the requirement of 3.5.5. The battery shall produce 9 volts dc.

4.5.6 Low battery indicator. A battery adapter shall be installed in the battery housing of the Assessor under test, and its cable connected to a power supply. The voltage shall be set to 9 volts as read on the multimeter. The low battery indicator will not be illuminated. The voltage shall then be reduced to 5.0. volts. Verify that the low battery indicator is illuminated.

4.5.7 Controls and switches. Conformance with the 3.5.7 requirement shall be demonstrated wearing arctic mittens, as indicated.

4.5.8 Power on indicator LED. Conformance with the requirement of 3.5.8 shall be demonstrated via manually operating the Assessor.

4.5.9 Optical instrument case. Demonstrate conformance with the requirement of 3.5.9 by placing all items into the case, as indicated.

4.5.10 Input current. During the Low Battery Indicator test of 4.5.6, the input current will be measured to verify that it does not exceed 5 milliamps.

4.5.11 Test modes. Ensure that the requirement of 3.5.11 has been met.

4.5.12 Light levels. Ensure that the requirement of 3.5.12 has been met.

4.5.13 Field of view. In order to test for the requirement of 3.5.13, the Assessor shall be mounted in a holding fixture on a precision rotary table with a high intensity light source illuminating the Assessor’s test port. A dioptometer shall be used to measure the field of view. The field should be no less than 30 degrees.

4.5.14 Reliability. The Assessor reliability requirement of 3.5.14 shall be demonstrated by a MTBF compliance of 10,000 hours. For guidance, the provisions of Test Plan 30-1, figure 20, of MIL-HDBK-781 are recommended. A test accept decision shall be based on demonstration of 1200 hours accumulated on each Assessor with no reliability failures. A test rejection decision shall be based on occurrence of one chargeable failure in 1200 hours for any test set. If the contractor performs the test for a longer period of time to allow one failure without rejection, the test set reliability requirement will be demonstrated by a MTBF compliance of 10,000 hours using the provisions of Test Plan 30-2, figure 20 of Handbook-781 as guidance. A Test accept decision shall be based on demonstration of 1960 hours accumulated on each test set with no more than one reliability failure.

4.5.15 Interchangeability. Ensure that all units, adapters and replaceable parts are completely interchangeable by performing a parts exchange using no less than five(s) disassembled Assessors. No modifications may be made to components to alleviate mating problems during this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.6 Environmental tests. Environmental testing may be performed in accordance with the recommended procedures in 63. However, the contractor may develop and use alternate test methods in a test plan showing equivalent methods following the test plans approval by the Government contracting office. When the contractor's test plan deviates from the recommended test procedures, the contractor shall be required to show where his own tests test methods for verifying the specified performance are equivalent. Also, measurements taken by the contractor during verification testing apparatus, e.g. environmental test chamber, vibration test table, pendulum shock device, or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented.

4.6.1 High temperature. Verify compliance with the requirement of 3.6.1. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.2 Low temperature. Verify compliance with the requirement of 3.6.2. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.3 Altitude. Verify compliance with the requirement of 3.6.3. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.4. Temperature shock. Verify compliance with the requirement of 3.6.4. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.5 Shock. Verify compliance with the requirement of 3.6.5. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.6 Vibration. Verify compliance with the requirement of 3.6.6. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.7 Humidity. Verify compliance with the requirement of 3.6.7. The test recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.8 Immersion. Verify compliance with the requirement of 3.6.8. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.9 Fungus. Verify compliance with the requirement of 3.6.9. The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constituted failure of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.6.10 Salt fog. Verify compliance with the requirement of 3.6.10. The tests prescribed in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.11 Sand and dust. Verify compliance with the requirement of 3.6.11 The tests recommended in 6.3, or equivalent validation method, may be used. Failure to meet the requirement shall constitute failure of the test.

4.7 Visual inspection. Each Assessor shall be visually examined for workmanship per 3.10 or any other nonconforming requirement such as design, finish, coating, construction, weight, marking or specific machine finishes and use of special lubricants for which specific tests are considered unnecessary. In such cases, visual examination using and this specification comprise a more comprehensive basis of evaluation for examining these requirements.

4.8 Electromagnetic interference (EMI). Ensure the Assessor conforms to the EMI test requirements delineated in Appendix A.

4.9 Inspection of packaging. Packaging shall be inspected for compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Assessor is intended for use as a test device to provide a Go/No-Go operational check of Night /vision Systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

c. Commercial standards may be used for environmental testing (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

d. Necessary actions by the contractor in the event of a first article test failure.

e. Quantity and schedule for first article testing for all line items.

f. Sampling for Group B and Group C testing unless otherwise specified, should be conducted per the requirements of ASTM E 122-89.

g. Actions required relative to Group C and Group D failures.

h. Disposition of test samples shall be in accordance with the contract on purchase order.

i. Packaging requirements.

6.3 Environmental inspection procedures. The test methods and procedures of MIL-STD-810 listed below are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the material to be tested.

Inspection	Spec. Paragraph	MIL-STD-810 method	MIL-STD-810 procedure	Remarks
High temperature (operating)	3.6.1	501.1	I	Operation temp (Step 4) is 52°C for 6 hours.
High temperature (storage)	3.6.1	501.1	I	Storage temp. (Step2) is 71°C for 3 hours.
Low temperature (operating, storage)	3.6.2	502.1	I	Operation temp. (Step 4) shall be -32°C for 9 hours. Storage temp (Step 2) shall be -62°C for 9 hours
Altitude	3.6.3	500.1	I	Operate altitude chamber @ 15K for operation and 40K for storage.
Temperature shock	3.6.4	503.1	II	Use temp. of 71°C to -62°C when non-operation
Shock	3.6.5	516.2	I	Use 40 g's @ 11 milliseconds.
Vibration	3.6.6	514.2	X	Use profile AR of table 514.2 VII and Figure 514.2-7.
Humidity	3.6.7	507.1	II	Use 10 days.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

Immersion	3.6.8	512.1	I	Use 30 minutes.
Fungus	3.6.9	508.1		
Salt fog	3.6.10	509.1		
Sand and dust	3.6.11	510.1	I	

6.4 Electromagnetic interference verification. The test set operational performance shall not be adversely affected during or after exposure to electromagnetic interference at the frequencies between 14.0 KHz and 40.0 GHz as measured per the emission and susceptibility requirements in MIL-STD-461. These requirements are:

a. RE102
b. RS102
c. RS103

6.5 Definitions. See 3.12.

6.6 Subject term (key word) listing.

ANVIS
Goggles, Night Vision
Image Intensifier

6.7 Changes form previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5805-0105)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

APPENDIX A

ELECTROMAGNETIC INTERFERENCE (EMI) VERIFICATION PROGRAM FOR TEST SET, ELECTRONIC SYSTEMS, TS-4348/UV

A.1. SCOPE

A.1.1 Scope. This appendix outlines and specifies the requirements for the electromagnetic interference (EMI) verification program applicable to Test Set, Electronic Systems. It is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 The EMI verification program for the Test Set, Electronic Systems shall be completed as per the following test plans:

a. The EMI Qualification Test Plan, to be performed on first article equipment as specified in the contract or purchase order.

b. The EMI Production Verification Test Plan, to be performed on samples of production equipment as specified in the contract or order.

A.2. APPLICABLE DOCUMENTS

Not applicable to this appendix.

A.3. TEST CONDITIONS

A.3.1 The equipment shall be set up in a shielded enclosure with the test equipment and its related antennas and sensing equipment.

A.3.2 The test shall be conducted at normal room ambient temperature of 20° C to 30°C (68°F to 86°F).

A.3.3 Power shall be provided to the equipment under test through line impedance stabilization networks (LISN) which provide a 50 ohm signal output port for the measurement of the radio frequency signal.

A.3.4 The equipment under test shall be accessible to the operator for switching and retuning as required.

A.4. VERIFICATION PROCEDURE

A.4.1 Radiated emissions, electric field, 2 MHz to 18 GHz. The equipment is to be tested for radiated emissions of the electric field over the frequency range of 2 MHz to 18 GHz using

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

the measurement setup of Figure A-1. The emissions shall not exceed the levels shown in Figure A-2. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields.

A.4.2 Radiated susceptibility, electric field, 10 KHz to 40 GHz.. The equipment is to be tested for radiated susceptibility to the electric field over the frequency range of 10 KHz to 40 GHz using the measurement setup of Figure A-3. The equipment shall not exhibit any malfunction, degradation of performance or deviation from specified indications, beyond the tolerances indicated in this specification, when subjected to the electric fields shown below in Table A-I. Up to 30 MHz, the limits shall be met for vertically polarized fields. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields. Circular polarized fields are not acceptable.

Table A-I. Radiated susceptibility limits - electric field.

Frequency range	Limit level (volts/meter)
10 KHz-2 MHz	20
2 MHz - 30 MHz	50
30 MHz - 1 GHz	50
1 GHz - 18 GHz	50
18 GHz - 40 GHz	50

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

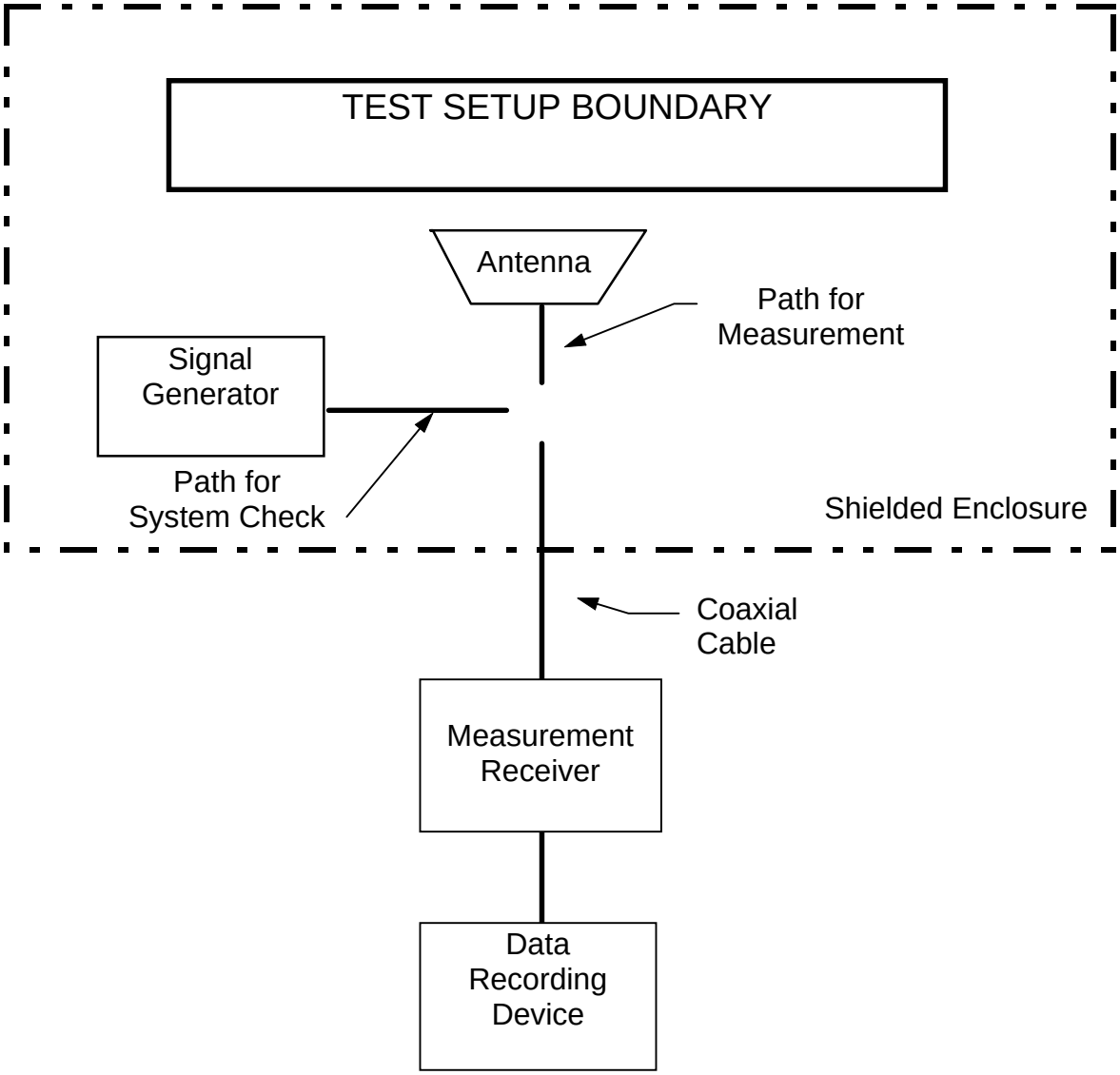


FIGURE A-1. Test setup, radiated emissions, electric field, 2 Mhz to 18 GHz.

(RE102)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

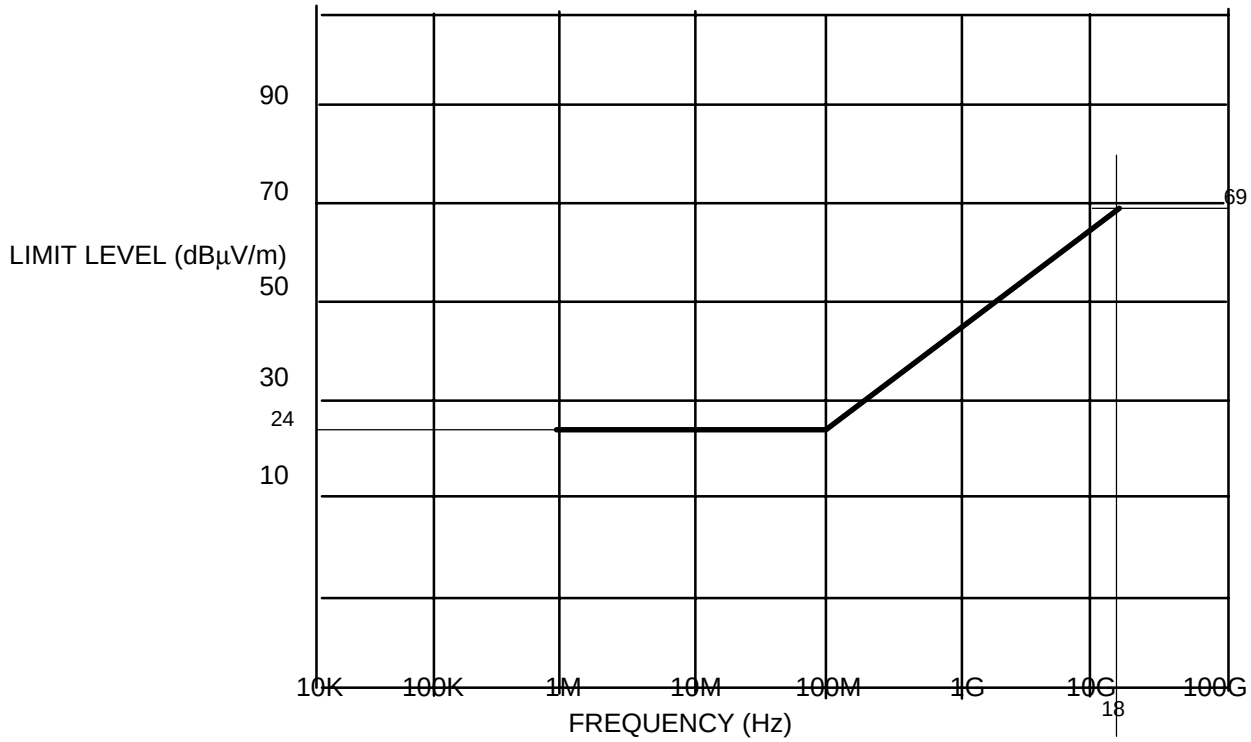


Figure A-2. Radiated Emission limit, electric field.

(RE102)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

APPENDIX A

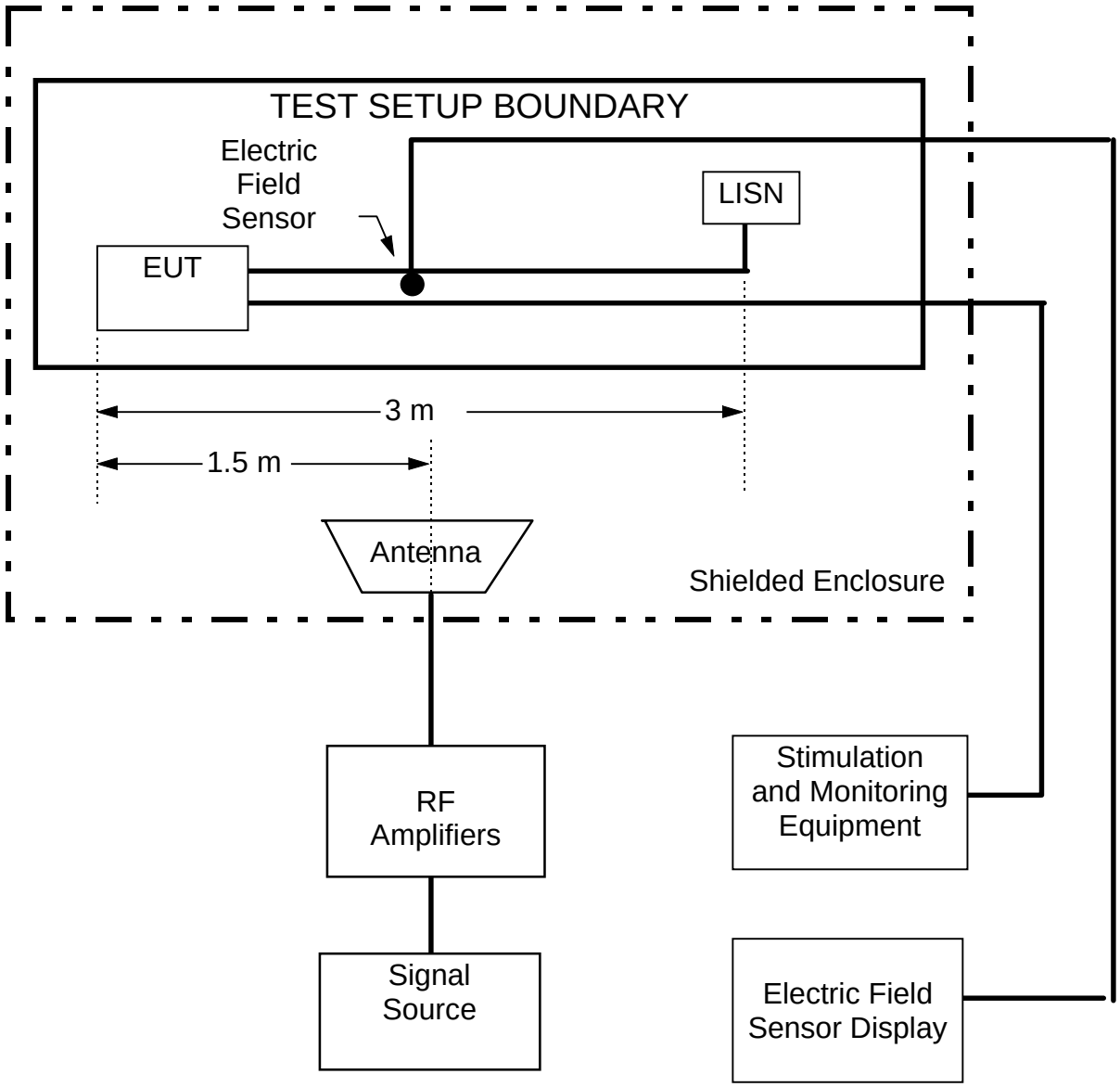


FIGURE A-3. Test setup, radiated susceptibility, electric field, 10 Mhz to 40 KHz.

(RS103)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49318A	**2. DOCUMENT DATE (YYMMDD)** 970519

3. DOCUMENT TITLE TEST SET, ELECTRONIC SYSTEMS, TS-4348/UV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE	b. TELEPHONE *Include Area Code)* (1) Commercial: (732) 532-2735 (2) AUTOVON: 992-2735
c. ADDRESS *(Include Zip Code)* US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, ATTN: AMSEL-LC-LEO-E-EP FORT MONMOUTH, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.